

Multiple Choice (10 marks)

1. When a wireless user authenticates to any AP, both of them go in the course of four-step authentication progression which is called _____
 - (a) AP-handshaking
 - (b) 4-way handshake
 - (c) 4-way connection
 - (d) wireless handshaking
2. Exploitation of the Heartbleed bug permits
 - (a) overwriting cryptographic keys in memory
 - (b) a kind of code injection
 - (c) a read outside bounds of a buffer
 - (d) a format string attack
3. A hash function guarantees the integrity of a message. It guarantees that the message has not be
 - (a) Replaced
 - (b) Overview
 - (c) Changed
 - (d) Violated
4. What was the first buffer overflow attack?
 - (a) Love Bug
 - (b) SQL Slammer
 - (c) Morris Worm
 - (d) Code Red
5. The _____ is anything which your search engine cannot search.
 - (a) Haunted web
 - (b) World Wide Web
 - (c) Surface web
 - (d) Deep Web

True / False (10 marks)

Answer True or False

6. True or False: A front-door is a security method that enables traceable access to computer systems and their information.
7. True or False: CBF is not a recognized block cipher operating mode in cryptography.
8. True or False: Wireless traffic sniffing is a process used for analyzing wireless traffic, aiding in forensic investigations and troubleshooting.
9. True or False: Encryption and decryption guarantee both confidentiality and integrity of the data.
10. True or False: A packet filter firewall filters data primarily at the application or transport layer of the OSI model.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to access dictionary key's element by index.
12. Write a function to calculate distance between two points using latitude and longitude.

End of Paper